AIORI HACKATHON 3 Internet Measurements Track

PROBLEM STATEMENT PS-032

RPKI-PathWatch

Path and Reachability Analysis During RPKI State Changes Changes

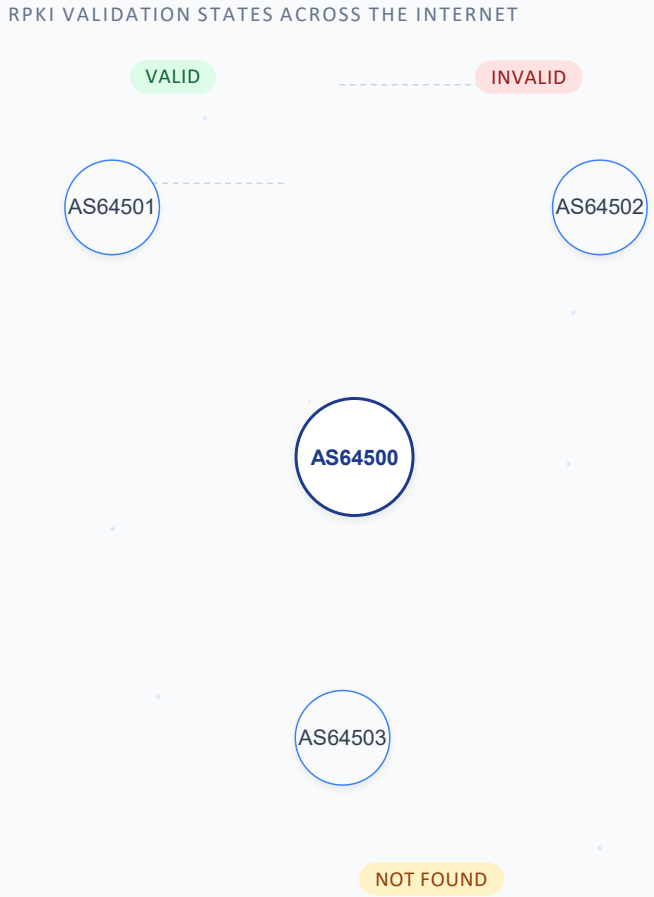
TEAM CODE

A3-PS032-TC538

Student Member 1: Akshat Dhabalia : akshat.bsmsfs7242914@nfsu.ac.in
Student Member 2: Krishn Dafda : krishn.btmtcs25@as.nfsu.edu.in
Faculty Member: Parthajit Borah

National Forensic Science University (NFSU), Guwahati Campus

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Abstract & Problem Description

Why RPKI validation-state changes matter to Internet routing

Abstract

Resource Public Key Infrastructure (RPKI) and Route Origin Validation (ROV) allow networks to cryptographically verify that a BGP route announcement is authorized for its originating Autonomous System (AS). When a validation state changes — for example, when a prefix-origin pair transitions from **Valid** to **Invalid** — the routing impact depends on which networks enforce ROV, what alternative paths exist, and the policies of upstream ASes. RPKI-PathWatch proposes an event-driven measurement workflow that identifies validation-state transitions, aligns them with BGP announcements, withdrawals, and AS-path changes observed at public route collectors, and reports before-and-after visibility across multiple vantage points. The expected contribution is a reproducible, evidence-based analysis pipeline that distinguishes temporal correlation from demonstrated causation and explicitly documents uncertainty in historical event analysis.

Approach: historical-data prototype first; live monitoring is a future extension, not a current not a current capability.

Problem Description

- RPKI/ROV provide cryptographic validation of BGP route origins — but adoption is partial across networks.
- Validation states are Valid, Invalid, and NotFound; each transition can change route preference.
- Routing impact is policy-dependent: a state change does not imply global unreachability.
- BGP behavior must be observed at multiple collectors (RIPE RIS, RouteViews) to distinguish local vs. widespread effects.
- Today, analysts cross-reference RPKI monitors and BGP tooling manually; correlation is hard to reproduce.
- Need: a workflow that aligns validation transitions with BGP events across collectors, with explicit before-and-after windows.

We make no claim about the frequency or severity of incidents — only about the need for a reproducible correlation workflow.

Background

Technical foundations needed to understand the problem

Autonomous Systems & BGP

- An Autonomous System (AS) is a routing domain under one administrative policy, identified policy, identified by an ASN.
- BGP exchanges announcements (reachable prefixes) and withdrawals (no longer reachable) between ASes.
- An AS-path is the sequence of ASNs a route has traversed — used for loop detection and route selection.

Prefixes & Route Origin Validation

- A prefix is an IP address block (e.g., 203.0.113.0/24) advertised in BGP.
- A Route Origin Authorization (ROA) binds a prefix to an authorized origin ASN, maximum prefix length, and validity interval.
- Route Origin Validation (ROV) classifies a route as **Valid**, **Invalid**, or **NotFound**.

Route Selection & Convergence

- Networks that enforce ROV may reject or de-prefer Invalid routes; others accept them unchanged.
- BGP convergence is the time taken for routing to stabilize after a change propagates.
- Outcome of a state change depends on local policies and available alternate paths.

Prefix Visibility from Multiple Collectors

- Public collectors (RIPE RIS, RouteViews) observe BGP from different vantage ASes.
- A route visible at one collector may be invisible at another — neither is 'the global view'.
- Multi-collector analysis is needed to characterize widespread vs. local changes.

ILLUSTRATIVE EXAMPLE — PREFIX 203.0.113.0/24, ORIGIN AS64500

VALID → INVALID *after ROA tightening or revocation*

Possible outcomes: route rejected by ROV-enforcing ASes; alternate path via another upstream selected; or no observable change at non-enforcing collectors. A transition does not automatically imply global unreachability.

Literature & RFC Review

Standards, measurement infrastructure, and prior work

[1] RFC 6811 — BGP Prefix Origin Validation

IETF · January 2013 · P. Mohapatra, J. Scudder, D. Ward, R. Bush, R. Austein

Establishes: The procedural standard for how a BGP speaker uses RPKI origin validation data validation data when selecting routes.

Relevance to PS-032: Defines the Valid/Invalid/NotFound decision logic our workflow must workflow must correlate against observed BGP behavior.

[2] RFC 8481 — Clarifications to BGP Origin Validation Based on RPKI

IETF · September 2018 · R. Bush

Establishes: Operational clarifications on how routers should treat Invalid routes and how and how origin validation interacts with route selection.

Relevance to PS-032: Explains why the same state transition can produce different routing routing outcomes across networks — central to our multi-collector analysis.

[3] RFC 9582 — A Profile for Route Origin Authorizations (ROAs)

IETF · 2024 · G. Huston, G. Michaelson, R. Loomans, T. Bruijnzeels

Establishes: The current ROA encoding format and validation rules; obsoletes RFC 6482.

Relevance to PS-032: Defines the structure of the authorization objects whose creation/revocation drives the state transitions we analyze.

[4] RFC 8210 — Version 1 of the RPKI to Router Protocol

IETF · September 2017 · R. Bush, R. Austein

Establishes: The protocol for distributing validated RPKI cache data from a trusted cache to cache to routers (RTR).

Relevance to PS-032: Underlies how validation states propagate to routers; informs how we informs how we interpret the timing of state changes in our correlation window. Version 2 is Version 2 is currently an IETF draft (draft-ietf-sidrops-8210bis).

[5] NIST RPKI Monitor

NIST · rpki-monitor.antd.nist.gov

Establishes: A long-running measurement platform that monitors global RPKI validation validation dynamics (valid, invalid, not-found originations).

[6] RIPE RIS & RouteViews

RIPE NCC · University of Oregon · public BGP collectors

Establishes: Two widely-used public BGP routing-data collection platforms providing multi-vantage observations of announcements, withdrawals, and AS-paths.

Positioning. RPKI monitoring (e.g., NIST) and BGP route analysis (e.g., RIPE RIS, RouteViews tooling) already exist as separate capabilities. RPKI-PathWatch proposes an **integrated, reproducible integrated, reproducible event-analysis workflow** that adds explicit before-and-after timelines and multi-collector comparison — not a claim that no existing tool performs these analyses. performs these analyses.

RPKI-PathWatch — Proposed Solution

Event-driven workflow correlating RPKI state changes with BGP behavior

WORKFLOW

1. Data Acquisition
2. Transition Detection
3. Event Correlation
4. Multi-Collector Analysis
5. Temporal Analysis
6. Visualization
7. Evaluation

EXPECTED OUTPUTS

State-Transition Event Timeline

Chronological log of Valid/Invalid/NotFound transitions with transitions with origin ASN and prefix.

Before/After AS-Path Comparison

Structured diff of AS-paths observed for the same prefix prefix before and after a transition.

Announcement & Withdrawal Summary

Counts and timestamps of BGP announcements and withdrawals in the event window.

Prefix Visibility by Collector

Per-collector visibility matrix showing where a prefix was was observed at each timestamp.

Observed Recovery Interval

Where measurable, the time between a transition and observed routing stabilization.

Evidence-Based Event Summary

Per-event report with confidence caveats and explicit non-non-causality statement.

MVP scope. The historical-data analysis prototype is the priority: pipeline, transition analysis, before-and-after comparison, dashboard. A live monitoring pipeline is a future extension, not a current capability.

Optimization Proposed

Conventional vs. RPKI-PathWatch analytical workflow

Analysis Dimension	Conventional / Separate Analysis	Proposed RPKI-PathWatch Approach	Expected Benefit
RPKI state & BGP event correlation	Manual cross-reference between RPKI monitors and BGP-looking-glass tools.	Automated alignment of validation-state transitions with BGP events by event timestamp.	Faster, reproducible correlation across many events.
Before/after AS-path comparison	Ad-hoc BGP dump diffs at single collector; output format varies by analyst.	Structured AS-path diff computed per transition event, normalized across collectors.	Comparable, reviewable evidence; citable per-event artifacts.
Multi-collector observations	Per-collector analysis in isolation; cross-collector reasoning is manual.	Unified view across RIPE RIS and RouteViews; per-collector visibility matrix.	Distinguish local vs. widespread routing changes.
Prefix visibility analysis	Limited to single-vantage looking-glass views.	Visibility matrix over all selected collectors across the event window.	Better characterization of reachability impact.
Temporal stabilization analysis	No standardized windowing; recovery intervals hand-estimated.	Defined before/after event windows with consistent timestamp alignment.	Repeatable recovery-interval estimation; comparable across events.
Reproducibility & evidence reporting	Notebooks and screenshots; hard to re-run or cite.	Pinned datasets + versioned pipeline + per-event report export.	Auditable, citable analysis; reviewable by third parties.

Important. Benefits are expected workflow improvements in interpretability and reproducibility — not measured performance gains. No accuracy, latency, or throughput numbers are claimed. numbers are claimed. Existing capabilities (NIST RPKI Monitor, RIPE RIS, RouteViews) remain foundational; RPKI-PathWatch integrates and structures their outputs.

Architecture Design (UML Diagram)

Component view — data sources, processing, storage, and output

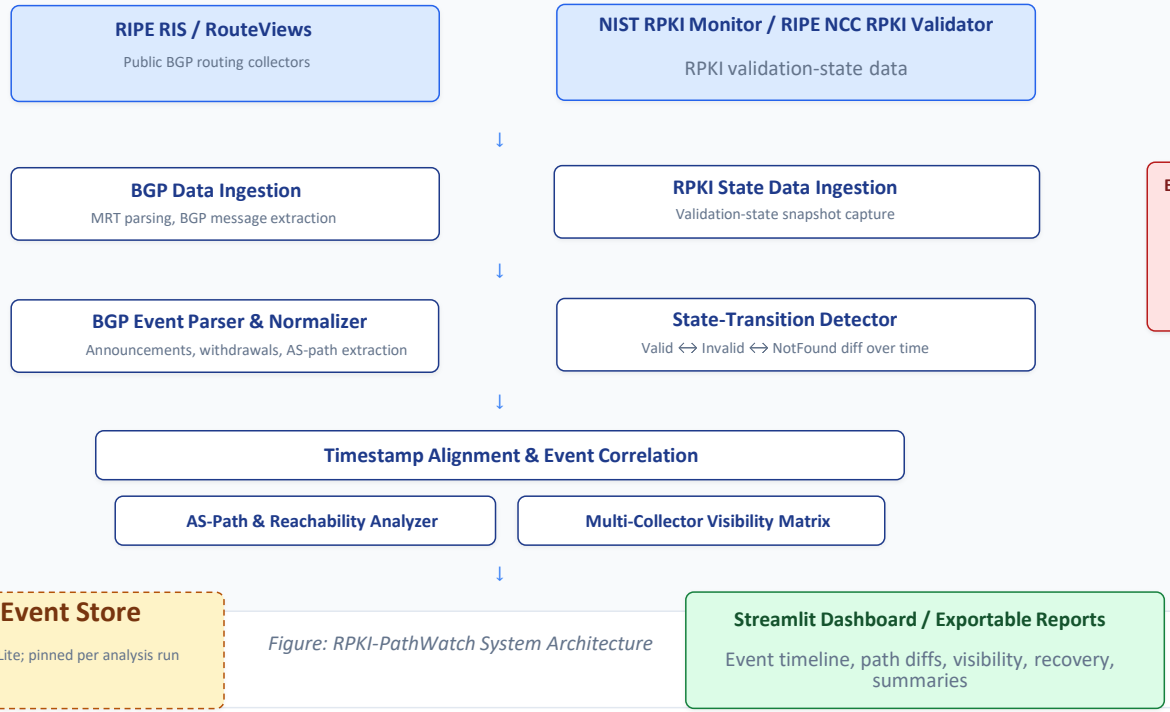


Figure: RPKI-PathWatch System Architecture

Data Flow Diagram

External entities · processes · data stores · outputs

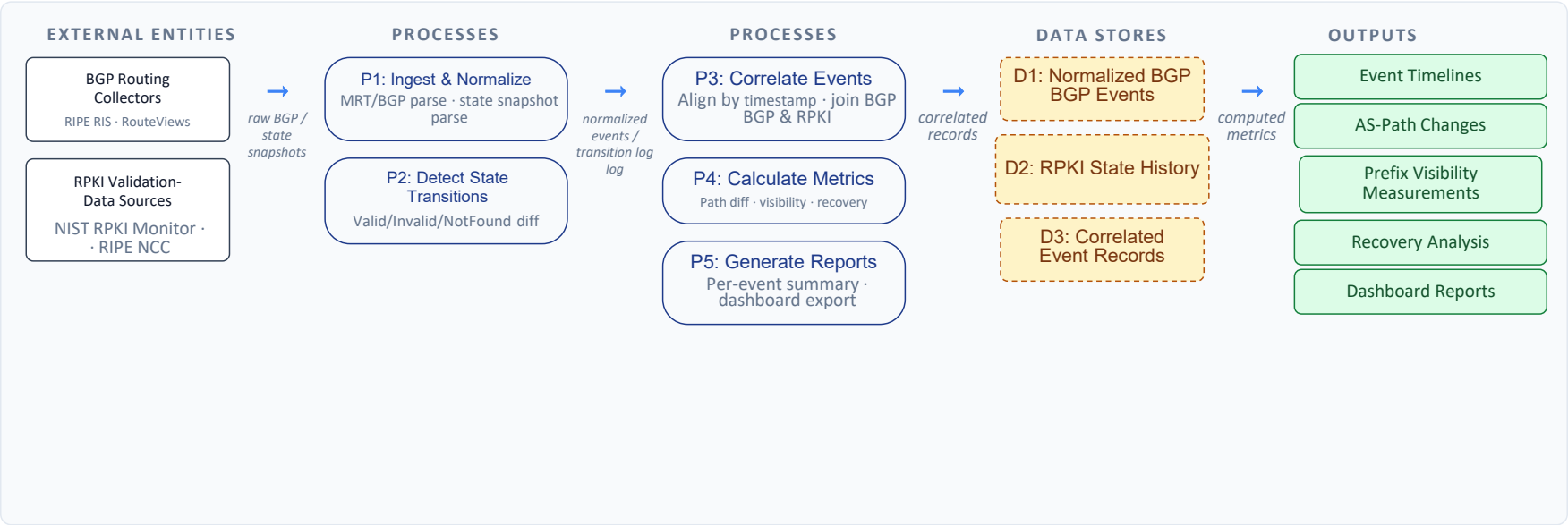


Figure: Data Flow Diagram (DFD)

■External Entity ◯Process ▭Data Store □Output

Deployment Timeline

Estimated milestone plan for a student research prototype

Total estimated duration: ~14 weeks. All durations are planning estimates for a student research prototype, not committed deadlines. Dependencies are sequential except where noted. where noted.

1 Problem Study, Standards Review & Dataset Selection — Est. 2 weeks

RFC 6811/8481/9582/8210 review; survey of NIST RPKI Monitor, RIPE RIS, RouteViews; select candidate historical events.

Deliverable: Annotated reading list + dataset shortlist with access notes.

2 Data-Source Inspection & Ingestion Pipeline — Est. 3 weeks

MRT/BGP parsing (e.g., libbgp/bgpkit); RPKI state snapshot capture; schema design for D1 & D2.

Deliverable: Working BGP/RPKI loaders writing to normalized event stores.

3 RPKI Transition Identification & BGP Event Correlation — Est. 3 weeks

State-transition detection logic; timestamp alignment; per-event correlation records (D3).

Deliverable: Transition-event log + correlation logic with unit tests.

4 Path, Withdrawal & Prefix-Visibility Metrics — Est. 2 weeks

AS-path diff; announcement/withdrawal counts; per-collector visibility matrix; recovery-interval estimator. interval estimator.

Deliverable: Metrics computation pipeline producing per-event JSON/CSV outputs.

5 Dashboard Development — Est. 2 weeks

Streamlit app; event timeline, path-diff view, visibility heatmap, recovery chart; per-event report export. report export.

Deliverable: Streamlit dashboard MVP with at least one validated historical event loaded.

6 Historical-Event Validation, Testing & Documentation — Est. 2 weeks

Run pipeline on 2–3 selected events; manual cross-validation against external reports; write up caveats and limitations. limitations.

Deliverable: Validation report + reproducible release (pinned datasets + versioned code).

Dependencies. Phase 2 depends on Phase 1 dataset choices. Phase 3 depends on Phase 2 loaders. Phase 4 depends on Phase 3 correlation records. Phases 5 and 6 depend on Phase 4 metrics. on Phase 4 metrics. Phases 5 and 6 may overlap by ~1 week.

Bibliography

Primary sources — RFC Editor, NIST, RIPE NCC, RouteViews

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2. [2] R. Bush. "Clarifications to BGP Origin Validation Based on Resource Public Key Infrastructure (RPKI)." RFC 8481, IETF, September 2018. https://www.rfc-editor.org/rfc/rfc8481
3. [3] G. Huston, G. Michaelson, R. Loomans, T. Bruijnzeels. "A Profile for Route Origin Authorizations (ROAs)." RFC 9582, IETF, 2024. https://www.rfc-editor.org/rfc/rfc9582
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Note on verification. All RFC titles and dates were verified against the RFC Editor (www.rfc-editor.org). RFC 9582's correct title is 'A Profile for Route Origin Authorizations (ROAs)', not 'RPKI to Authorizations (ROAs)', not 'RPKI to Router Protocol, Version 2' — the latter is currently an IETF draft (draft-ietf-sidrops-8210bis). URLs are official sources; no fabricated DOIs or authors. fabricated DOIs or authors.

AIORI HACKATHON 3

Thank You for Your Attention!

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Problem Statement PS-032 · Internet Measurements Track

Student Member 1: Akshat Dhabalia : akshat.bsmsfs.7242914@nfsu.ac.in

akshat.bsmsfs.7242914@nfsu.ac.in

Student Member 2: Krishn Dafda : krishn.btmtcs25@as.nfsu.edu.in

Faculty Mentor: Parthajit Borah

National Forensic Science University (NFSU), Guwahati Campus

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